Sexually Transmitted Infections (STI) & Pelvic Inflammatory Disease (PID)

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Block: Endocrine/Urology/ Reproduction

Thread: Infection & Immunity

Conflict of Interest: Dr. Weber has no conflicts of interest to disclose

Learning Objectives

By exam time, you should be able to do the following:

1. Describe the major bacterial causes of STIs and their microbiologic characteristics.
2. Differentiate the clinical presentations of gonorrhea and chlamydia and describe their major complications.
3. Compare the stages of syphilis and describe their clinical manifestations.
4. Differentiate the pathogens and clinical manifestations of the major causes of genital ulceration.
5. Describe first-line treatments for gonorrhea, chlamydia, and syphilis and recall their mechanisms of action and major adverse effects.

Further Reading

2021 Sexually Transmitted Diseases Guidelines www.cdc.gov

2024 Sexually Transmitted Disease Surveillance www.cdc.gov

National STI Curriculum https://www.std.uw.edu/

Current Medical Diagnosis & Treatment 2018 Eds. Maxine A. Papadakis, et al. New York, NY: McGraw-Hill. http://accessmedicine.mhmedical.com/content.aspx?bookid=2192&sectionid=167993948

Clinical Infectious Disease 2015 Eds. David Scholossberg. Cambridge University Press.

Study Questions

1. An 18-year-old patient comes in to the student clinic for evaluation. She requests testing for sexually transmitted infections. How do you evaluate her?

2. A 22-year-old medical student who recently returned from a medical volunteer trip to Haiti presents to you for dysuria and penile discharge. While on his trip, he ate local street food, swam in the ocean, and drank alcohol. He had unprotected sexual intercourse with a new female partner. He denies any fever, chills, or lymphadenopathy. On exam, he appears nontoxic and age-appropriate. There is thin, watery fluid spontaneously draining from his urethra. There is no evidence of genital ulcerations or local lymphadenopathy. What would your next steps be for diagnosis and treatment?

3. A 23-year-old patient presents to the Emergency Department with lower abdominal pain and fever. She is sexually active with multiple partners and does not always use protection. She reports normal menses about 1 week ago. On exam, she is febrile to 101F and appears moderately uncomfortable. There is diffuse tenderness with palpation of the lower abdomen. Pelvic exam reveals cervical motion tenderness and mucoid discharge. What are the complications of this syndrome if left untreated?

4. A 26-year-old patient with HIV on medications comes into clinic with a diffuse, pruritic rash that involves his trunk, extremities, palms, and soles. He reports having intercourse with a new male partner about 6 weeks ago. He does not specifically recall any penile lesions or drainage. On exam, he is mildly uncomfortable but not toxic. There is a diffuse maculopapular rash over his chest, arms, and hands. Oral membranes reveal whitish patches. Cervical lymphadenopathy is present. What is your diagnosis?

5. A healthy 21-year-old patient comes in to the student clinic with 2 days of a painful blister on his penis. He reports recent unprotected intercourse with a new male partner this week. He denies any fever or chills. On exam, he is anxious appearing but nontoxic. Genital exam is notable for several small vesicles on the glans penis. There is no urethral discharge. There is mild inguinal lymphadenopathy. What is on your differential diagnosis?

A. Definition

1. STI is an infection that is passed from one person to another through sexual contact via vaginal, oral, and anal sex. Rarely can STIs be transmitted from close skin contact.

2. Examples:

a. This includes but is not limited to:

- HIV (separate lecture!)
- Syphilis caused by *Treponema pallidum*
- Gonorrhea caused by *Neisseria gonorrhoae*
- Chlamydia caused by *Chlamydia trachomatis*
- Herpes Simplex Virus (HSV)
- Human Papilloma Virus (HPV)
- Trichomonasis caused by *Trichomonas vaginalis*
- Other genital ulcerative diseases
- NOT Mpox

B. Assessing Your Patient

1. History

a. Sexual history is a component of comprehensive, patient-centered care

b. Obtain a sexual history in an **open, nonjudgmental manner**

c. Identify sexual orientation and gender identity

Examples of questions to consider when assessing your patient:

- Any prior history of STIs
- Number of lifetime sexual partners
- Gender(s) of partners
- Use of protection/contraception

- Sites of sexual contact
- Use of drugs or sex for drugs/pay

2. Physical

a. anogenital exam in all patients

b. gynecological exam if appropriate

c. skin and mucous membranes

d. lymph nodes

C. Chlamydia

1.Caused by the bacteria *Chlamydia trachomatis*

a. Very small gram-negative bacteria that lacks a peptidoglycan wall

b. The elementary body, a small, infectious, but nonreplicating particle found in secretions, attaches to and enters a host cell, such as an endocervical or urethral columnar epithelial cell.

c. Within 8 hours, the now-intracellular elementary body interacts with glycogen and transforms into a reticulate body, which begins to multiply within an isolated intracellular structure referred to as an inclusion. The reticulate bodies are the noninfectious replicating form.

d. Within 48 hours, some reticulate bodies begin to reorganize back to elementary bodies.

e. Within 72 hours, most of the reticulate bodies have transitioned back to elementary bodies, and the inclusion either undergoes lysis at the host cell wall or the intact inclusion (containing numerous elementary bodies) is released into the extracellular space, a process called extrusion.

f. Regardless of whether the inclusion undergoes lysis or extrusion, the elementary bodies are released to infect adjacent cells or to be transmitted to and infect another person.

2. Clinical Syndromes

a. **Nongonococcal urethritis** is the most common manifestation in patients. It can be described as pain with urination and thin to thick mucoid discharge from the urethra

b. Cervicitis is described as inflammation of the cervix, often with a red appearance on pelvic exam with yellow mucopurulent discharge.

c. Infection can **ascend the genital tract** and cause additional syndromes such as epididymitis, orchitis, prostatitis, acute salpingitis, and PID.

d. Other sites **outside of the genital tract** can be infected including: pharyngitis (oral sex), proctitis (anal sex), conjunctivitis (Trachoma), perihepatitis (Fitz-Hugh-Curtis Syndrome), and reactive arthritis (triad of conjunctivitis, urethritis, and arthritis)

e. **Patients can also be asymptomatic, especially those assigned female sex at birth, which leads to delay in diagnosis/treatment, continued transmission of infection, and increases the risk of developing PID**.

3. Diagnosis

a. Nucleic Acid Amplification Test (**NAAT**) is the preferred test for diagnosing *Chlamydia trachomatis* infections.

b. Test can be run on vaginal fluid, urine, urethral swab, oropharyngeal swab, rectal swab

c. Endocervical swabs and urethral swabs (men) can be sent for culture.

d. Test patients who have **symptoms concerning for STI or who have tested positive for another STI**.

e. Preemptive screening for high risk patients, biologic females, pregnant patients, and patients who practice anal sex.

4. Treatment

a. **Doxycycline 100 mg twice daily x 7 days for urethritis/cervicitis**

b. **Alternative: azithromycin 1g po x1 or levofloxacin 500 mg po daily x 7 days**

c. **Also treat empirically for gonorrhea with ceftriaxone** intramuscular 500 mg x1 dose if results not back

d. No resistance has been described

e. Test of cure in 3 weeks if symptomatic/pregnant/suboptimal treatment, or retest in 3 months in everyone else

D. Gonorrhea

1. Caused by the gram-negative bacteria *Neisseria gonorrhoeae*.
    a. Replicates by binary fission, leading to the diplococci form
    b. Pili attach to host cells allowing cause of disease
    c. These pili are encoded by genes that undergo recombination with changing Aas, making the pili a challenging target for vaccination
    d. Lipo-oligo saccharide (LOS) exterior also help evade the immune system

2. Clinical Syndromes
    a. Similar to chlamydia, the most common manifestations of gonococcal infection include **urethritis** and **cervicitis**. It is often described as painful urination with purulent discharge.
    b. Infection can **ascend the genital tract** and cause addition symptoms such as epididymitis, orchitis, prostatitis, acute salpingitis, and PID.
    c. Other **sites outside the genital tract can be infected** including: pharyngitis, proctitis, bacteremia and septic arthritis.
    d. Disseminated gonococcus is rare (<3%) but can occur. It is characterized by malaise, fever, polyarthralgias, tenosynovitis, and dermatitis. Cultures and clinical suspicion are crucial for diagnosis
    e. Patients can also be asymptomatic, especially biologic female sex, which leads to delay in diagnosis/treatment, continued transmission of infection, and increases the risk of developing PID.

3. Diagnosis
    a. Nucleic Acid Amplification Test (**NAAT**) is the preferred test for diagnosing *Neisseria gonorrhoeae* infections.
    b. Test can be run on vaginal fluid, **urine**, urethral swab, oropharyngeal swab, rectal swab
    c. Endocervical swabs and urethral swabs can be sent for culture.
    d. Test patients who have symptoms concerning for STI or who have tested positive for another STI.
    e. Preemptive screening for high risk patients, biologic females, pregnant patients, MSM

4. Treatment
    a. **Ceftriaxone intramuscular 500 mg x1 dose**
    b. **Also treat for chlamydia**, even if infection is ruled out with doxycycline 100 mg bid x 7 days
    c. Alternate if allergy to cephalosporins: gentamicin 240 mg IM x1 PLUS azithromycin 2g oral x1
    d. **Resistance has been described** and is progressing
    e. Test of cure in 3 weeks if symptomatic/pregnant/suboptimal treatment, or retest in 3 months in everyone else

E. Pelvic Inflammatory Disease (PID)

1. Definition: **infection of higher up structures of the female reproductive tract** including uterus (endometritis), fallopian tubes (salpingitis) and/or ovaries (oophoritis)

2. Main pathogens:
    a. Chlamydia trachomatis
    b. Neisseria gonorrhoeae

3. Clinical Manifestations
    a. Fever
    b. Lower abdominal pain
    c. Abnormal menstrual bleeding
    d. **Cervical motion tenderness** "chandelier sign"

4. **Complications**
    a. Scarring of the fallopian tubes leading to **sterility or ectopic pregnancies**
    b. Abscesses
    c. Peritonitis, or infection of the peritoneal lining
    d. **Perihepatitis** (Fitz-Hugh-Curtis Syndrome), which is characterized by right upper quadrant pain due to inflammation of the liver capsule and peritoneum.

5. Diagnosis
    a. **Often difficult to diagnose**
    b. Suspect in sexually active patients with a uterus and ovaries with lower abdominal pain without alternative etiology and one of the following: cervical motion tenderness, uterine tenderness, adnexal tenderness
    c. Pelvic inflammatory disease should be in the differential for transgender men with a uterus and fallopian tubes who have vaginal intercourse

6. Treatment
    a. Parenteral or IM ceftriaxone plus doxycycline with or without metronidazole x 14 days

F. Syphilis

1.Caused by the corkscrew-shaped gram-negative organism (spirochete) *Treponema pallidum*.

    a. It is very small and does not grow well on culture

2. Clinical syndromes can be categorized by stages of infection:
    a. **Primary syphilis** is characterized by one or more painless ulcers, known as a chancre.
    b. **Secondary syphilis** is characterized by dissemination of the organism and typically occurs 2-6 weeks after primary syphilis. Symptoms may include a diffuse maculopapular rash, lymphadenopathy, mucosal patches, alopecia, and condyloma lata. The primary lesion may still be present.
    c. **Tertiary syphilis** is rare, but may follow untreated infection after 15-30 years. It includes end-organ manifestations of the cardiovascular system (aortitis, coronary artery disease), skeletal system (osteitis), and neurologic system (tabes dorsalis, paresis, dementia, strokes) among others.

3. Some patients will be **asymptomatic**, but may test positive for syphilis. When this occurs, patients are described as having **latent syphilis**. This is divided into early latent and late latent stages.

a. **Early latent syphilis** is diagnosed when infection occurs within preceding 12 months.

b. **Late latent syphilis** occurs when the timing of initial infection is unknown or acquisition is expected to be greater than the preceding 12 months.

c. **It is important to distinguish between these 2 stages as the treatment differs**.

4. Diagnosis

a. Unable to culture *Treponema pallidum*, so diagnosis relies on indirect testing including:

b. **Non-treponemal tests** include rapid plasma reagin (RPR) or venereal Disease Research Laboratory (VDRL), which measure antibodies against lipid antigens that are released by the host in reaction to *Treponema spp*.

c. **Treponemal-specific tests** include fluorescent treponemal antibody absorption (FTA-ABS) or microhemagglutination assay for Treponema pallidum (MHA-TP)

d. Traditional diagnostic testing would include a non-treponemal test (RPR) followed by confirmatory treponemal-specific test (FTA-ABS)

e. **The current recommendation is for reverse algorithm**: first perform a treponemal-specific testing followed by non-treponemal test

f. The RPR results are reported as a serologic titer. Normal response to treatment is four-fold drop in titer within 3 months and should be followed to confirm success of treatment and to evaluate for new infection.

5. Treatment

a. Primary/ Secondary/ Early Latent infection is treated with **penicillin** G intramuscular 2.4 million units x **1 dose**

b. Tertiary/ Late Latent infection is treated with penicillin G intramuscular 2.4 million units x **3 weekly doses**

c. If penicillin allergic, doxycycline can be used as alternate therapy.

d. Neuro and ocular syphilis is treated with **continuous infusion** of aqueous penicillin G x14 days

e. Jarisch-Herxheimer Reaction: occurs during treatment as a result of inflammation related to dying spirochetes. Characterized by flu-like syndrome with fever, myalgias, and headache. Typically self-limiting, though has been known to cause premature labor.

6. Congenital syphilis occurs when a fetus is infected in utero with *Treponema pallidum*.

   a. Signs of early congenital disease include "snuffles", rash, lymphadenopathy, and osteitis

   b. Signs of late congenital syphilis include neurosyphilis (especially deafness), bony abnormalities (saddle nose, saber shins, Hutchinson's teeth), and keratitis

   c. Congenital syphilis also increases the neonatal mortality rate

G. Herpes Simplex Virus (HSV)

1. Viral infection that most commonly manifests as small cluster or group of vesicles in the oral and/or genital regions. It can also be asymptomatic. Generally, HSV-1 is felt to be associated with oral infection and HSV-2 with genital infections, however genital HSV-1 infections are increasing.

2. Infections can be **primary**, meaning it occurs in a person with no antibodies, or **recurrent** as reaction of latent virus in a person with seropositivity. Seropositive persons may shed the virus without showing signs or symptoms of infection.

3. Epidemiology

   a. >85% of adults have serologic evidence of HSV-1 exposure

   b. ~ 25% of adults have serologic evidence of HSV-2 exposure

   c. **Age** increases the risk of HSV-1 and HSV-2 seropositivity

   d. **Number of sexual partners** increases the risk of HSV-2 seropositivity

   e. **HSV-2 increases the risk of HIV acquisition**

4. Clinical Manifestations:

   a. small vesicular lesions that erupt over 1-2 days, often **painful**

   b. skin lesions (buttocks, thighs, finger "herpetic whitlow", wrestlers "herpes gladiatorum")

   c. Ocular- keratitis, keratoconjunctivitis, blepharitis

   d. Neurological- encephalitis, meningitis (recurrent "Mollaret"), Bell's palsy

   e. GI- esophagitis, especially in immunocompromised patients or those with

HIV/AIDS, fulminant hepatitis

f. Disseminated infection

g. Erythema multiforme

5. Diagnosis

a. Clinical- do not test asymptomatic patients!

b. **PCR** of fluid from unroofed vesicle

c. Viral culture, sensitivity ~50%

d. Serology- pregnancy to screen

e. Tzanck preparation showing intranuclear inclusion bodies and multinucleated giant cells

f. Keratitis diagnosed by appearance of dendritic ulcers on fluorescein stain

g. Encephalitis/meningitis pcr

h. GI- histology

6. Treatment

a. Initiating treatment within 72 hours of an outbreak can decrease the severity and length of the illness

b. Acyclovir 400 mg 3x daily or valacyclovir 1g daily by mouth for 7-10 days

c. Patient initiated therapy for recurrences with short courses is reasonable

H. Other Genital Ulcerative Diseases

1. Lymphogranuloma venereum (LGV)

a. Caused by *Chlamydia trachomatis* serotypes L1-L3

b. Clinically characterized by single **painless ulcer** with associated **painful lymphadenopathy**

c. Can also cause proctitis

d. Diagnosis is typically via NAAT

e. Treatment is usually with doxycycline or azithromycin x3 weeks

2. Granuloma inguinale
    a. Caused by *Klebsiella granulomatis*
    b. Also known as "Donovanosis"
    c. Clinically characterized by **painless, destructive lesion with beefy red borders** and high vascularity
    d. Diagnosis is made by biopsy of lesions
    e. Treatment is usually with doxycycline or azithromycin until lesion resolution

3. Chancroid
    a. Caused by *Haemophilus ducreyi*.
    b. Characterized by **painful, ragged appearing ulcer** often with "kissing" lesion on inner thigh and **painful lymphadenopathy**
    c. Diagnosis is based on culture and clinical suspicion
    d. Treatment is with oral azithromycin 1g x1 dose or IM ceftriaxone 250 mg x1 dose

I. Human Papilloma Virus (HPV)

1. **Most common STI**
    a. Estimated annual incidence of 14 million cases/year in the US
    b. Prevalence estimated around 80 million in the US
    c. Peak prevalence is during first 10 years of sexually activity, **between ages 15-25**, however most infections resolve spontaneously within 6-12 months

2. Many strains of HPV, with about 40 infecting the mucosal epithelial cells (genital, oral, pharyngeal)
    a. Types **6 and 11 cause genital warts**, known as condylomata
    b. Types **16 and 18 cause high grade dysplasia and SCC of the cervix, vagina, vulva, and penis**
    c. Linked to approximately 12,000 cases/year of cervical cancer, which is the 3rd most common cause of cancer in biologic females
    d. Additionally, type 16 among others have been associated with SCC of the

oropharynx as it can be transmitted through open mouth kissing and oral sex.

3. Diagnosis

a. Areas of high grade dysplasia may appear as white patches when exposed to acetic acid during exam

b. **Cytologic exam** via pap smear, colposcopy and biopsy

c. **HPV DNA testing** is often added to improve sensitivity

4. Treatment/Prevention

a. Majority of infections will clear spontaneously

b. There is no antiviral treatment available

c. Vaccination for all patients ages 11-12 (can start at age 9preferentially with the 9-valent (6, 11, 16, 18, 31, 33, 45, 52, 58)

d. Vaccine can also be given to most persons after age 13- 26 as a "catch up"

e. Vaccination for adults ages 27-45 can be considered in discussion with their physician.

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. *Answer: Whenever a patient comes in requesting STI testing, more information is needed. It is important to obtain a detailed sexual health history to determine the likely exposures and risks that may warrant further testing for a patient. Be open and nonjudgmental when obtaining these intimate details. Specific questions to ask may include the following: do you have any specific concerns? Have you or your partner(s) been diagnosed with an STI? What symptoms are you experiencing and for how long? How many current and lifetime partners do you have? Do you have sex with men, women, or both? Do you participate in oral, vaginal, and/or anal sex? Do you use protection with sexual acts? How often? Have you ever tested positive or been treated for an STI? Are drugs or alcohol present or a part of sexual acts?*

   After obtaining a good history, the physical exam is next. Assess the patient's oral membranes, lymph nodes, abdomen, and genital/perineal regions. Look for discharge from the urethra, genital ulcers, and inguinal lymphadenopathy. A pelvic exam is recommended in females for inspection visual inspection of the vagina and cervix, as well as to assist with specimen collection if needed.

2. *Answer: This patient is presenting with classic symptoms of urethritis, namely dysuria and urethral discharge. The history of recent unprotected sexual intercourse with a new partner further heightens suspicion for STI. The diagnostic test of choice would be a urine or urethral swab for Chlamydia trachomatis and Neisseria gonorrhoeae NAAT. Empiric treatment should be offered and includes doxycycline 100 mg twice daily x 7 days AND intramuscular ceftriaxone 500 mg.*

3. *Answer: This young patient is presenting with symptoms/signs of pelvic inflammatory disease, notably fever, lower abdominal pain, and cervical motion tenderness. If left untreated, PID can be complicated by abscesses (tubo-ovarian), perihepatitis, peritonitis, scarring of the female genital tract, ectopic pregnancies, and infertility.*

4. *Answer: Syphilis has been called the "great imitator" as it can present with a variety of symptoms and signs that may delay diagnosis. This patient has secondary syphilis, as evidenced by a diffuse maculopapular rash that is often pruritic and covers the palms/soles, lymphadenopathy which can often be diffuse, and oral mucous membrane patches. Other features of secondary syphilis include alopecia and condyloma lata. Sometimes the initial chancre can still be present on exam.*

a. *In contrast, primary syphilis is characterized by a painless ulcer called a chancre. It is often described as looking "punched out" with heaped up, smooth borders. The lesion occurs at the site of inoculation by Treponema pallidum and often resolves on its own. It can occur in obscure sites and may be missed on initial evaluation.*
b. *Tertiary syphilis occurs many decades after primary infection if untreated. Patients may have nonspecific signs within the cardiovascular, neurologic, and skeletal system.*

5. *Answer: The best way to approach the differential for genital ulcers is to separate them into "painful" and painless" categories. For this patient, he is presenting with painful lesions and local lymphadenopathy. The 2 infections that can produce painful lesions include HSV (1 and 2) and chancroid (can also be painless). Other genital ulcerative diseases are painless, including primary syphilis, lymphogranuloma venerum (LGV), and granuloma inguinale. In these latter infections, typically the ulcer is painless but there can be associated painful lymphadenopathy. HSV is classically described as groups of vesicles clustered together. The chancre of syphilis is often well demarcated with a "punched out" appearance. The ulcer for LGV can begin as a raised papule followed by a shallow ulcer. Chancroid is rare in the United States, but is often characterized by a ragged, red ulcer with yellow base. This patient likely has HSV.*

Vaginitis

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Block: Endocrine/Urology/Reproduction

Thread: Infection and Immunity

Conflict of Interest: Dr. Cullen has no conflicts of interest to disclose

Further Reading

https://www.std.uw.edu/go/comprehensive-study/vaginitis/core-concept/all

ACOG Practice Bulletin Number 215, "Vaginitis in Nonpregnant Patients"

AAFP "Vaginitis: Diagnosis and Treatment": https://www.aafp.org/pubs/afp/issues/2018/0301/p321.html

Learning Objectives

By exam time, you should be able to do the following:

1. Develop a differential diagnosis for a patient with a chief complaint of "vaginal discharge"
2. Describe the microbiology and diagnostic findings (including microscopy) of yeast vaginitis, bacterial vaginosis, and trichomoniasis
3. Differentiate these infections by their clinical presentation and physical exam findings
4. Describe the key epidemiologic features and standard treatments for each

Study Questions

1. A 24yo patient presents malodorous vaginal discharge. Which of the following would help to differentiate between the two main infectious causes of this symptom?

a. Increased pH of vaginal discharge

b. Presence of many PMNs on microscopy

c. Presence of thin discharge

d. Pseudohyphae on microscopy

2. A 47yo patient presents with severe vaginal and vulvar itching and clumpy vaginal discharge. She denies medical history but acknowledges that she doesn't regularly seek medical care. Which of the following is **not** a risk factor for the infection most likely causing her symptoms?

a. Chronic systemic steroid use

b. Condom use

c. Poorly controlled diabetes mellitus

d. Recent antibiotic use

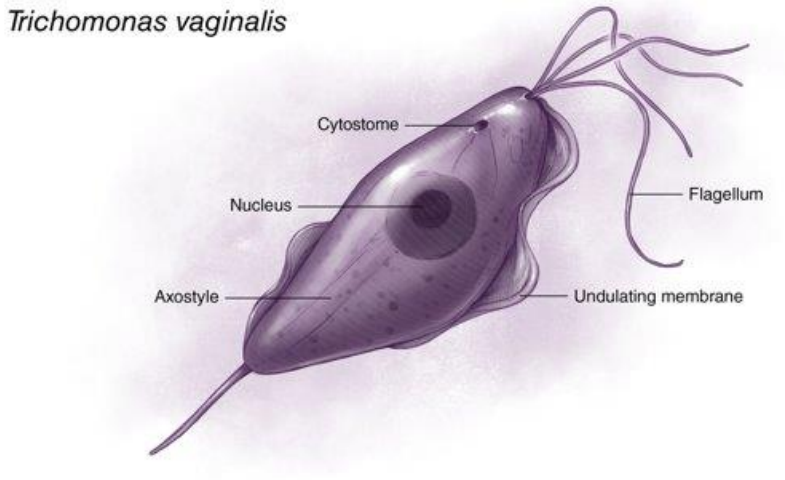
Differential Diagnosis for Vaginal Discharge

- STIs: gonorrhea, chlamydia, **trichomonas**, herpes simplex virus (especially primary vulvovaginal HSV2)
- **Bacterial vaginosis (BV)**
- **Vaginal candidiasis ("yeast infection")**
- Retained foreign body
- Desquamative inflammatory vaginitis (DIV)- etiology not entirely clear- may be autoimmune, hormonally related, or allergic- results in discharge, itching, swelling, erythema and pain.
- Structural lesion- you will learn about these in other sessions
  - Prolapsing polyp or fibroid
  - Necrosing intracavitary fibroid
  - Cervical cancer

Trichomoniasis

- Microbiology:
  - Single-celled flagellated anaerobic protozoan parasite *Trichomonas vaginalis*
  - Only known protozoan parasite that infects the genital tract
  - Four anterior flagella and one posterior flagellum embedded in the "undulating membrane" ~> jerky movement under the microscope
  - Unlike other protozoa (specifically flagellates), *T. vaginalis* has no cyst form, and its trophozoites cannot survive outside the host, thus *T. vaginalis* is only transmitted directly from person to person via sexual contact

        - The **trophozoite** is the active, growing, feeding stage in the protozoan life cycle
        - A protozoan cyst form is dormant and protected; **T. vaginalis has no cyst form**
    - If asymptomatic, may persist for months to years in the vagina



- Microscopic findings
- Clinical presentation and physical exam findings
    - Symptoms
        - Yellow-green vaginal discharge
        - Vaginal (+/- vulvar) pruritus
        - May be asymptomatic
    - Physical examination
        - **Frothy** yellow-green discharge
        - "Strawberry cervix"- cervical punctate hemorrhages, only in 2-5% of cases but is strongly suggestive of TV when present
        - Chronic infection may be associated with minimal vaginal discharge, mild pruritus, and/or dyspareunia

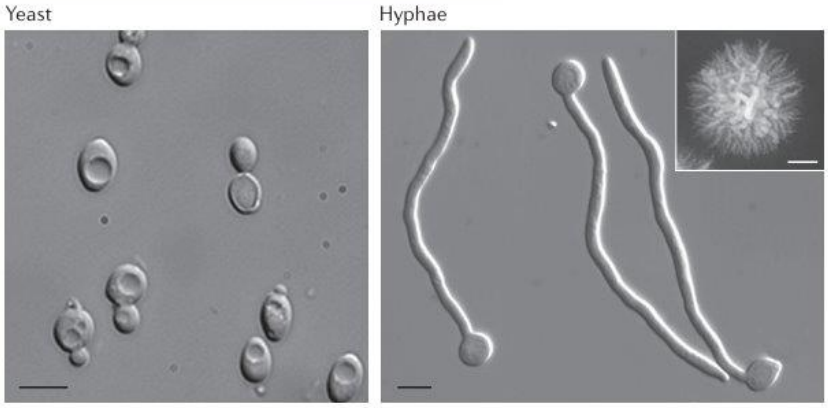


- Epidemiology
    - Almost always sexually transmitted (no dormant cyst form)
    - Infection associated with increased risk of acquiring HIV, as well as PID, infertility and cervical cancer
    - ~ 3.7 million people have trichomoniasis in the US
    - The **prevalence** of *T. vaginalis* infection among people with vaginas of reproductive age in the US is ~2%
    - 1.1 million new cases occurring each year
    - Men who have sex with men **appear to be** at low risk of acquiring trichomoniasis (**not screened**)
        - CDC screening recommendation does not include men unless HIV+
    - Transmission of *T. vaginalis* between women who have sex with women is common
- Diagnosis
    - Microscopy: motile trophozoite and many PMNs
        - Only motile for 10-20 min after collection, so most easily identified if slide is made and viewed shortly after swab collected
    - Vaginal pH elevated (5-6)
    - Nucleic Acid Amplification Testing (NAAT)
    - Culture: less accurate than NAAT (sensitivity 43-95%)
    -
- Treatment
    - Metronidazole 500mg twice a day x 7 days
        - Until 2018, dosing recommendation was 2g once- this is less effective than twice daily x 7 days, especially for people with vaginas
        - Metronidazole is an antibacterial (anaerobic coverage) and antiprotozoal medication
    - Alternative: tinidazole 2g once daily for 7 days
    - This is an STI, so partner treatment needed to prevent re-infection
    - From the CDC: "Because of the high rate of reinfection among women treated for trichomoniasis, retesting for *T. vaginalis* is recommended for all sexually active women approximately 3 months after initial treatment regardless of whether they believe their sex partners were treated (*137,1115*). If retesting at 3 months is not possible, clinicians should retest whenever persons next seek medical care <12 months after initial treatment.
        - Do not retest until **at least** 3 weeks after treatment completion- high likelihood of remnant trichomonads

Yeast Vaginitis

- Microbiology
    - About 90% is Candida albicans
    - The rest are “non-albicans”, including C. glabrata, krusei, tropicalis, parapsilosis, etc
- Microscopic findings
    - Candida albicans can be seen microscopically as hyphae, pseudohyphae, and/or budding forms
    - Non-albican strains (such as C. glabrata) can be seen as almost all budding forms





    - Hyphae are long, tube-like and without constrictions
    - Pseudohyphae have constrictions at sites of septation and are wider
    - Budding yeast are unicellular

- Clinical presentation and physical exam findings
    - Clumpy/cottage cheese like discharge



    - Vulvar and/or vaginal itching
    - External rash and irritation- especially in poorly controlled diabetes and immune suppressed individuals



- Epidemiology
    - Candida species are part of the normal (asymptomatic) vaginal flora in up to ¼ of people
    - Risk factors for symptomatic candida include recent antibiotic use, immune suppression, and diabetes
    - Candida is the 2nd most common cause of abnormal vaginal discharge, with about 1/3 of vaginitis cases seeking care being caused by candida
    - Difficult to study epidemiology due to easily available OTC treatments
    - Surveys show that up to 55% of people with vaginas have been dx with yeast at some point
- Diagnosis
    - Microscopy: Hyphae, pseudohyphae, budding yeast, many WBC (PMNs)
    - Vaginal pH is normal (~4.5)
    - NAAT

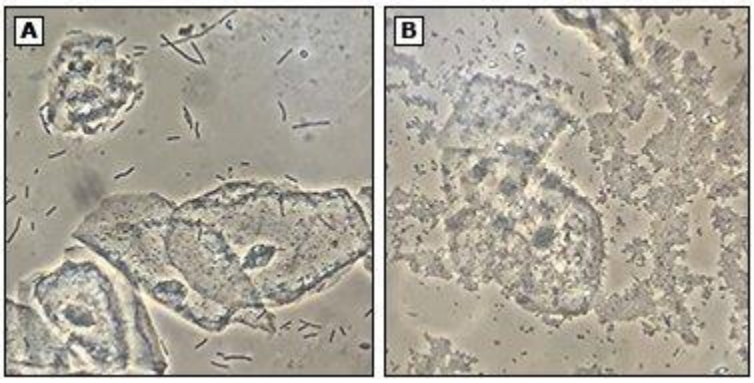
    - Yeast culture- should not be used for all cases- Culture without clinical correlation will overestimate prevalence due to asymptomatic colonization
        - Used for recurrent or persistent infections
        - Can differentiate species to evaluate for best treatment options
- Treatment
    - No need to treat if asymptomatic (for example, noted on pap result)
    - Uncomplicated: infrequent, non-pregnant, immunocompetent
        - Oral Fluconazole 150mg once
        - Topical Azoles- prefer 7d, can use 3d
        - Boric Acid- vaginal suppository x 7 days
    - Complicated
        - Pregnancy: 7-day topical azoles (clotrimazole and miconazole first line), may need 10–14-day treatment
            - We avoid oral fluconazole during pregnancy, particularly in the first trimester, because it may increase the risk of miscarriage and its impact on congenital anomalies is unclear
        - Immunocompromise: Fluconazole oral 150mg Q72h for 2-3 doses, or 14-day course of topical azole
        - Non-albicans: Vaginal boric acid 600mg vaginally for 14-21 nights

Bacterial vaginosis

- Microbiology
    - Shift from Lactobacillus predominance to bacterial diversity
        - Increase in facultative anaerobes primarily, especially gram-negative rods
        - Common species: Gardnerella vaginalis, Mycoplasma hominis, Ureaplasma urealyticum, Mobiluncus, Megaspera, Prevotella species, Bacteroides species, Atopobium vaginae etc
    - Anaerobes make proteolytic carboxylase enzymes-->break down vaginal peptides into volatile amines (malodorous & cause increased discharge and squamous cell exfoliation)
    - Gardnerella adheres to the squamous cells and creates a biofilm, which becomes a scaffold for other bacteria to adhere to
    - Why does the bacterial shift happen?
        - Wiping out/killing of Lactobacillus
            - Douching

- Cleaning inside the vagina with soap or other products (vaginal cleansers)
- Prolonged submerging in warm/hot water
- Certain sex partners? Exposure to BV associated pathogens

- Microscopic findings



A- normal, B- bacterial vaginosis (clue cells)

- Clinical presentation and physical exam findings
  - **Not inflammatory, thus "vaginosis" (as opposed to "vaginitis")**
  - Symptoms: vaginal discharge, odor, +/- mild irritation
  - Speculum exam: Thin, malodorous, off-white discharge
- Epidemiology
  - Makes up 40-50% of abnormal vaginal discharge visits
  - Prevalence is 29-50% in the general population of people with vaginas (self-swab, including both symptomatic and asymptomatic people)
  - Increases risk of preterm birth, post-birth infection, infections after vaginal surgery, acquisition and transmission of HIV and other STIs, development of PID, and persistence of HPV
- Diagnosis
  - Clinical Diagnosis: Amsel Criteria (need at least 3 of 4 to make diagnosis)
    - Thin homogenous discharge
    - Microscopy: Clue cells (at least 20%)
    - PH >4.5 (due to increased production of volatile amines by anaerobic bacteria and decreased production of lactic acid and hydrogen peroxide by lactobacilli)
    - Fishy odor when mixed with 10% potassium hydroxide (+ whiff-amine test)
  - Laboratory testing

        - NAAT (various FDA approved and lab-developed tests, testing for bacteria common in BV)
        - Gram stain (gold standard, but mostly research now, minimal clinical utility)
    - Testing choice guided by clinical experience/training, and availability of microscope
- Treatment
    - Metronidazole 500mg BID x 7 days **OR**
    - Metronidazole 0.75% gel vaginally QHS 5 days
    - Partner treatment????
        - Newer data suggest that it significantly decreases recurrence
    - Vulvovaginal care adaptations
        - Use unscented soap externally only
        - Avoid products inside the vagina (no douching!!)
        - Avoid scented products (tampons, pads, deodorizers)
        - Avoid tight pants or prolonged wet clothing
        - Use cotton underwear, avoid thongs, sleep without underwear

Vaginitis/vaginosis overview

Infection	PH	Microscopy	Symptoms	Exam
Candidal vaginitis (yeast)	*Normal (3.8-4.5)	**KOH or wet mount- budding forms and/or hyphae, increased PMNs	Clumpy discharge, itching, irritation, vulvar/groin rash	Thick, clumpy, adherent, white/yellow
Bacterial vagin**osis** (BV)	High (>4.5) **WHY?**	Clue cells Few to no lactobacilli	Discharge, fishy/garbage-like odor, irritation	Thin, grey/off-white, malodorous, homogenous
Trichomonas vaginalis	High (5-6)	Motile trichomonads, many PMNs	Thin discharge, burning, itching or irritation	**Frothy,** thin, malodorous, yellow-green, strawberry cervix (2-5% of cases)

- * normal vaginal pH is ACIDIC! Due to lactic acid and hydrogen peroxide production by lactobacilli
- ** KOH- 10% potassium hydroxide

Answers to Study Questions

1. A 24-year-old patient presents malodorous vaginal discharge. Which of the following would help to differentiate between the two main infectious causes of this symptom?

a. Increased pH of vaginal discharge

b. Presence of many PMNs on microscopy

c. Presence of thin discharge

d. Pseudohyphae on microscopy

Explanation: The two most common vaginal infections causing malodorous discharge are trichomoniasis and bacterial vaginosis. Both cause an increased pH of the vagina and present with a thin discharge. Pseudohyphae are seen with yeast vaginitis, not TV or BV. BV is a vaginosis and is non-inflammatory, so BV alone should not cause a significant increase in PMNs. TV is inflammatory and you should see many PMNs on microscopy.

2. A 47-year-old patient presents with severe vaginal and vulvar itching and clumpy vaginal discharge. She denies medical history but acknowledges that she doesn't regularly seek medical care. Which of the following is **not** a risk factor for the infection most likely causing her symptoms?

a. Chronic systemic steroid use

b. Condom use

c. Poorly controlled diabetes mellitus

d. Recent antibiotic use

Explanation: Severe vulvovaginal itching and clumpy discharge is most consistent with a yeast (candidal) infection. Condom use can be protective against yeast infections, while all other above options increase the risk of symptomatic yeast infections.